

MAGNUS DIERKING

Born in Germany, 21 September 1998

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WORK EXPERIENCE

	since 05/2026	PhD Student & Research Associate
PEARL		Computer Science Dept. · Technical University of Darmstadt Research in Robot Learning at the Interactive Robot Perception & Learning (PEARL) Group, supervised by Prof. Georgia Chalvatzaki.
	10/2024– 05/2025	Research Intern
Huawei R&D		Huawei Technologies Research & Development UK Limited · London, Shenzhen Extracurricular internship during Master studies. Working on reinforcement learning, imitation learning and supporting middleware for robotics as part of the Embodied AI team.
	2023–2024, 2025–2026	Research Assistant
IAS		Intelligent Autonomous Systems Group · Technical University of Darmstadt Developed a Python/C++ software stack for FR3 robotic arms, built on ROS2 Jazzy. Assisted Ph.D. researchers with experiments, visualization and studies on supporting theory in optimal transport, Bayesian inference and optimization for robot learning.
	2018–2021, 2022–2023	Working Student
mas		Medical Airport Service GmbH · Frankfurt Support personnel for the rescue station on the apron of Frankfurt International Airport.
	2021–2022	Teaching Assistant
TEMF		Institute for Accelerator Science and Electromagnetic Fields · Technical University of Darmstadt Developed lesson plans, theoretical / programming tutorials and provided mentorship and feedback for undergraduate students. Courses: Electrodynamics, Numerics for Electromagnetic Field Simulation

PUBLICATIONS

	2025	Ark
arXiv		Open-source, Python-first robotics framework that provides a Gym-style interface for collecting data, training policies, and switching seamlessly between simulation and real-robot deployment. It includes reusable modules for control, SLAM, motion planning and visualization, and integrates natively with ROS to accelerate end-to-end robotics research. First Author
	2025	OpenPyro-A1
IEEE RA-L		Open-source, low-cost bimanual half-humanoid robot designed for advanced manipulation research. It features a modular, repairable hardware design and supports coordinated two-handed tasks such as folding, cutting, and assembling. The platform enables teleoperation via a Meta Quest 3 and provides interfaces for learning-based controllers to support scalable, real-world robotics experimentation. Co-Author

EDUCATION

10/2022–
04/2026 Computational Engineering

Master of Science

GPA: 1.00 · Top of Class · Technical University of Darmstadt

Thesis: *Domain Randomized Deployment of Massively Parallel MPC*

Developed real-to-sim-to-real pipeline for massively parallel SMPC, implementing JAX-based planning stack with GPU-accelerated MuJoCo MJX and ROS2 control on Franka Research 3 for contact-rich manipulation and simulation studies. Evaluated algorithms up to 10 Hz with robust sim-to-real transfer. Analyzed online domain randomization limits, favoring contact-initiation parameters.

Advisor: Prof. Jan PETERS · Supervision: Dr. João CARVALHO, Dr. An Thai LE · Grade: 1.00

Robot Learning · Reinforcement Learning · Deep Generative Models · Intelligent Robotic Manipulation · Parallel Programming

Differential & Riemannian Geometry · Numerical Linear Algebra · Geometric Algebra · Information Theory 1-2 · Convex Optimization · Graph Signal Processing · Control Theory · Optimal Transport

10/2019–
10/2022 Computational Engineering

Bachelor of Science

GPA: 1.51 · Top 10% · Technical University of Darmstadt

Thesis: *Parallel Solution of Linear Systems Arising in Domain Decomposition Methods*

C++ implementation of a parallel solver for large-scale surface PDEs within an Isogeometric Analysis (IGA) framework.

Advisor: Prof. Sebastian SCHÖPS · Supervision: MSc. Maximilian NOLTE · Grade: 1.00

Robotics · Software Engineering · Algorithms and Data Structures · Functional & Object-oriented Programming · Geometric Modelling

Partial Differential Equations · Numerical and Statistical Methods · Mechanics 1-3 · Signals and Systems · Electrodynamics 1-2 · Maths 1-4

10/2018–
10/2019 Mechanical Engineering

Completed Coursework · Technical University of Kaiserslautern

Material Science 1-2 · Material Science Lab · Mechanical Design Project · Production Engineering · Business Organization

2018 Elisabeth-Langgässer Gymnasium

A-Levels

Math · English · History

TECHNICAL SKILLS

OS LINUX · Mostly Ubuntu, some Arch

Programming PYTHON · JAX, PyTorch, CVX, NumPy, SciPy, Pandas
C++ · Eigen, OpenMP
Basics Java, HTML, JavaScript, CSS

Simulation MuJoCo, PyBULLET

Tools GIT, ROS2, Docker, L^AT_EX

Hardware Franka Research 3, Trossen Viper, OpenPyro-A1
OptiTrack, Intel RealSense

OTHER INFORMATION

Scholarships 2024 · Erasmus Placements Program
2023 · Deutschlandstipendium

Languages GERMAN · Mother tongue
ENGLISH · Fluent
FRENCH · Basics

May 28, 2026